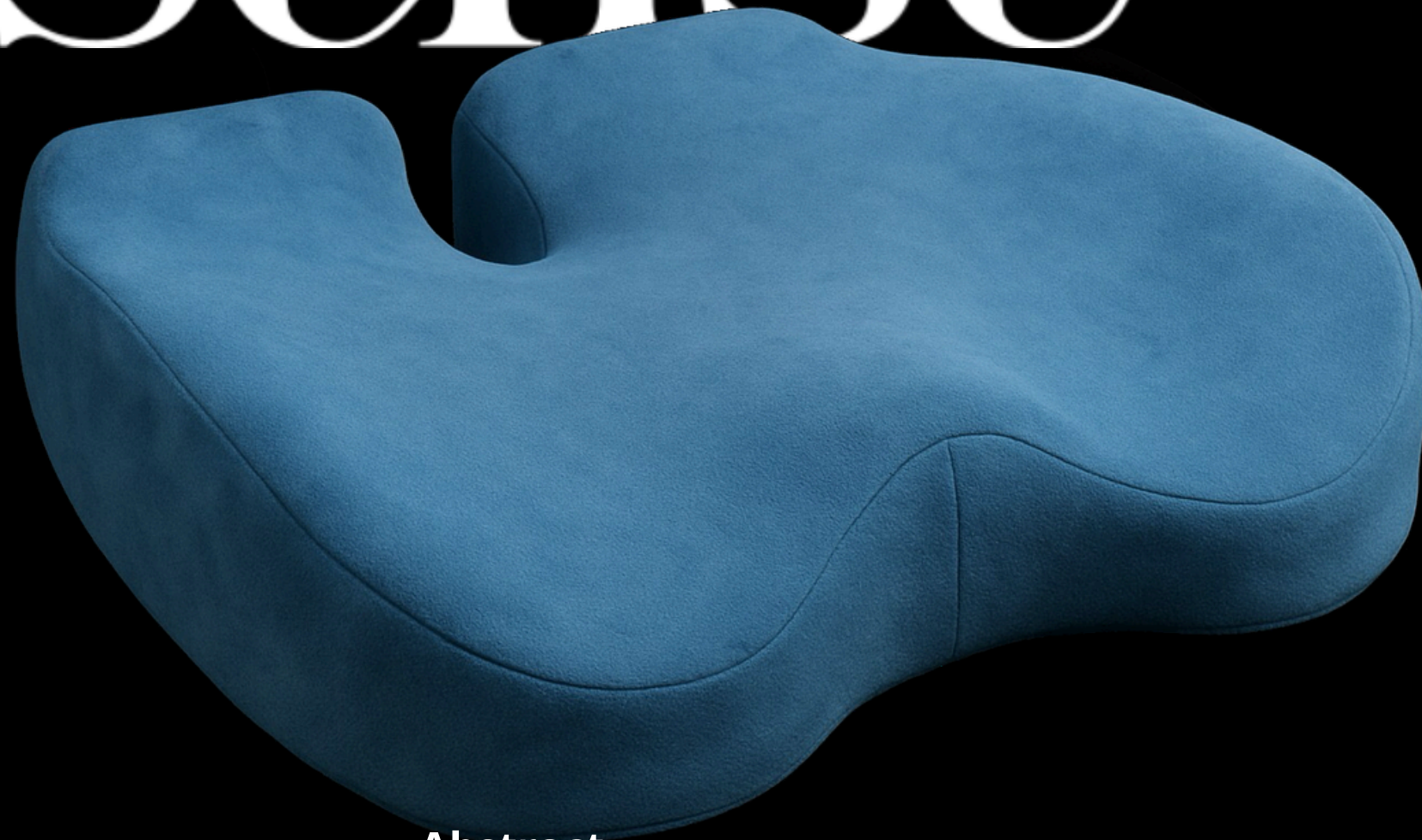


SitSense



Abstract

Prolonged sedentary behavior contributes to discomfort and reduced productivity. This project introduces **SitSense**, an integrated system designed to encourage micro-exercise breaks through behavioral nudges. The system combines an ergonomic cushion with a pressure sensor, vibration motors and a software interface that monitors sitting time. SitSense prompts the user with a gentle physical prompt when a sitting limit is exceeded, while intelligently respecting the user's schedule by referencing their calendar. The design is based on behavioral change models. Evaluation confirmed high user willingness to adopt the product, validating the potential of combining interactive technology and ergonomics to encourage healthier daily habits.

BACKGROUND RESEARCH

The modern environment, marked by prolonged sedentary behaviour and high cognitive demands, has become a major public health concern. Insufficient physical activity contributes to a wide spectrum of problems, including obesity, cardiovascular disease, musculoskeletal discomfort (particularly back pain), and elevated stress levels. These consequences extend beyond individual well-being, imposing substantial economic burdens through reduced productivity, presence and absence.

Micro-Breaks as a Preventive Strategy

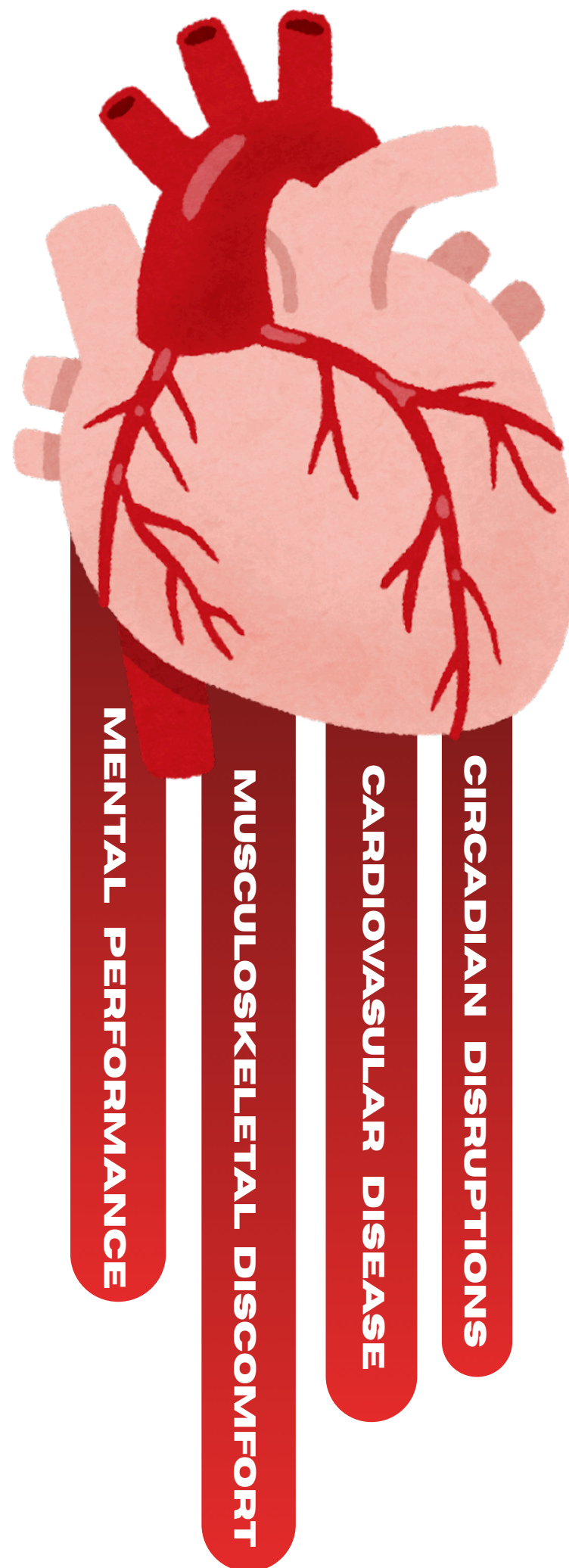
A growing body of research highlights the potential of micro-break activities as an effective intervention within both professional and educational settings. Short, movement-based breaks, such as stretching, walking, or performing light mobility exercises, have been shown to improve circulation, reduce fatigue, and mitigate musculoskeletal strain. Moreover, these brief physical activities enhance cognitive performance and alleviate mental fatigue, indicating a dual benefit for both body and mind. (Albulescu et al., 2022)

In Workplace Settings:

Sedentary work and prolonged sitting are established risk factors for declining health and productivity. A 2022 study published in Scientific Reports demonstrated that incorporating micro-exercise routines in the workplace effectively reduces long-term sickness absence (LTSA). When performed collectively, these brief activities support both physical well-being (e.g., reducing pain) and psychosocial health, offering a holistic approach to workplace wellness. (Radwan et al., 2022). Another study from 2023 (Impact of active breaks on sedentary behaviour and perception of productivity in office workers) highlights how reduction of sedentary behaviour (influenced by encouraged activity breaks) decreases the perceived stress, sleepiness, and simultaneously increases quality of life among productivity. (Batista-Ferreira et al., 2024)

In Educational Contexts:

Studies within educational environments show that short classroom activity breaks significantly improve students' physical activity levels, attention, and on-task behaviour. These "active breaks," which introduce brief changes in posture and movement, counteract the effects of passive breaks such as phone use, which tend to impair concentration and engagement. Movement breaks were proven to enhance academic performance, as well as increase alertness, concentration and enjoyment of university students.
ref. international journal of environmental research and public health



BEHAVIOUR CHANGE, NUDGING - METHODS

Nudging methods and strategies

facilitating – encourage people to pursue certain actions (that align with their personal values) – people tend to more often take action in the default options: by suggesting certain activities to partake in, the app is capable of making users do actually do it;

Facilitating, according to the paper, is also possible by positioning. In that instance, by visual arrangement, it is possible to suggest the recommended behaviour (in user interface for example).

Reinforcing is a nudging category that makes the nudges more memorable for the user. Just in time prompts, such as a notification/message after a prolonged sitting period, tune into the users' timeline, therefore being more effective.

App features on Self Determination Theory – motivations

Self Determination Theory states that humans have different types of motivations as well as different amounts of them. Autonomy could be setting goals in the app, customizable reminders/notification messages. Competence could be added by giving activity feedback and progress tracking – this provides positive performance feedback and sense of accomplishment. Relatedness provides group sharing/competition, but is not tightly related to this project.

Behaviour change models.

Fogg Behaviour Change model (ref) claims the human behaviour to be a product of factors: motivation, ability and triggers. According to that, the more motivation and ability the person has, the more they are likely to change their behaviour. Without the triggers, a behaviour cannot occur. It can be anything like a message, alarm or announcement. The article mentions: "successful triggers have three characteristics: First, we notice the trigger. Second, we associate the trigger with a target behaviour. Third, the trigger happens when we are both motivated and able to perform the behaviour". Triggers should be well-timed - with the highest motivation when the user has the ability to perform a behaviour.

Well timed triggers are accepted by the user. Therefore, the key factor is wanting to change ones elf's behaviour in the first place.

Behaviour change models

Fogg Behaviour Change model (ref) claims the human behaviour to be a product of factors: motivation, ability and triggers. According to that, the more motivation and ability the person has, the more they are likely to change their behaviour. Without the triggers, a behaviour cannot occur. It can be anything like a message, alarm or announcement. The article mentions: "successful triggers have three characteristics: First, we notice the trigger. Second, we associate the trigger with a target behaviour. Third, the trigger happens when we are both motivated and able to perform the behaviour". Triggers should be well-timed - with the highest motivation when the user has the ability to perform a behaviour.

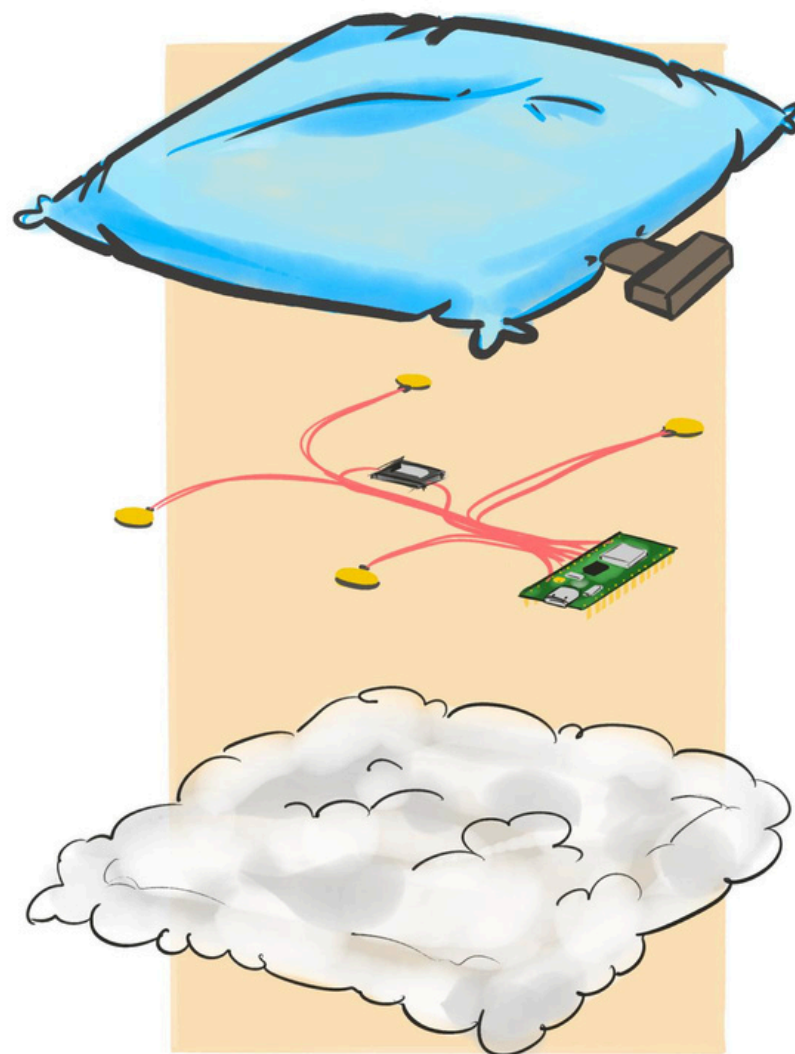
Well timed triggers are accepted by the user. Therefore, the key factor is wanting to change oneself's behaviour in the first place.

DESIGN JUSTIFICATION

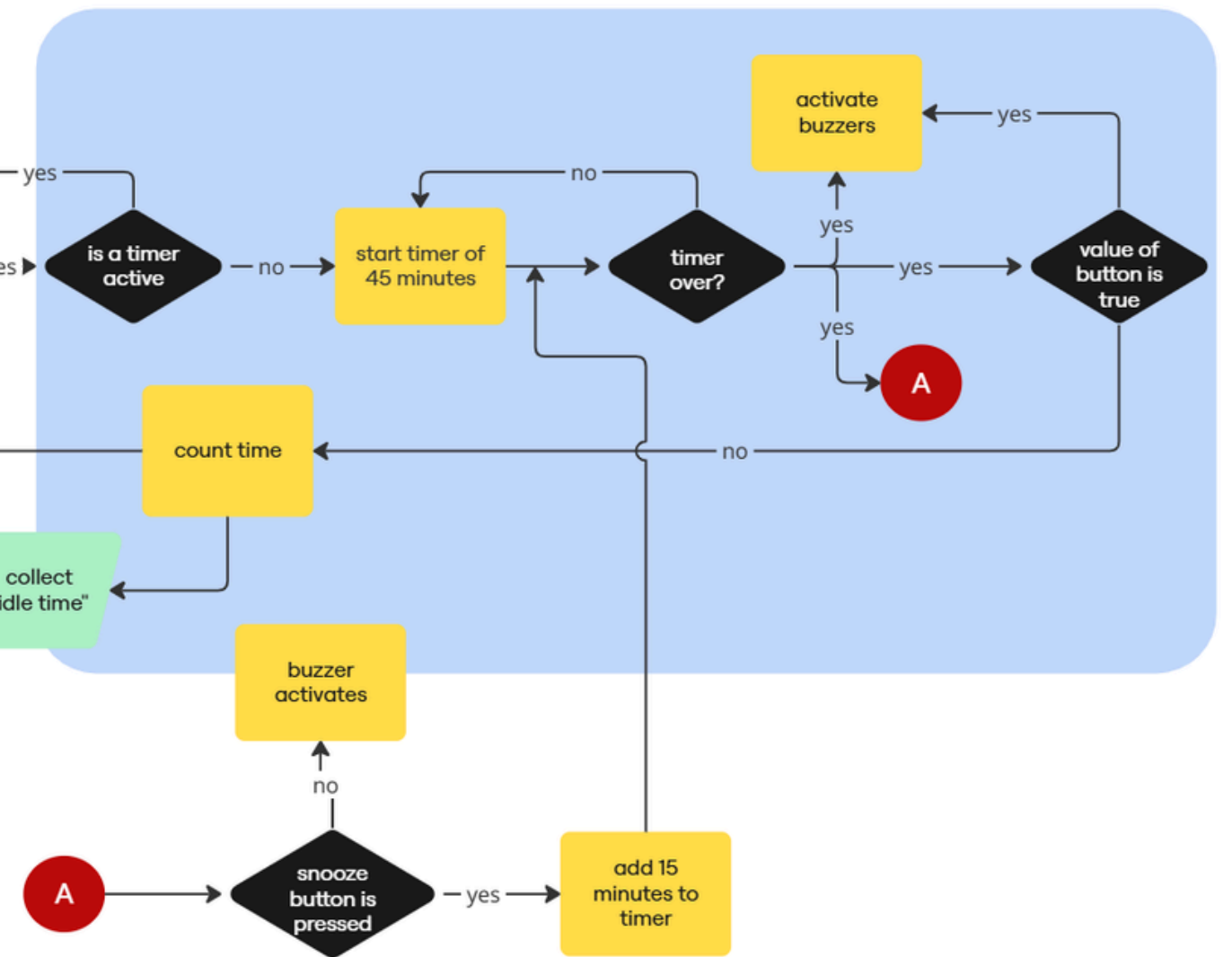
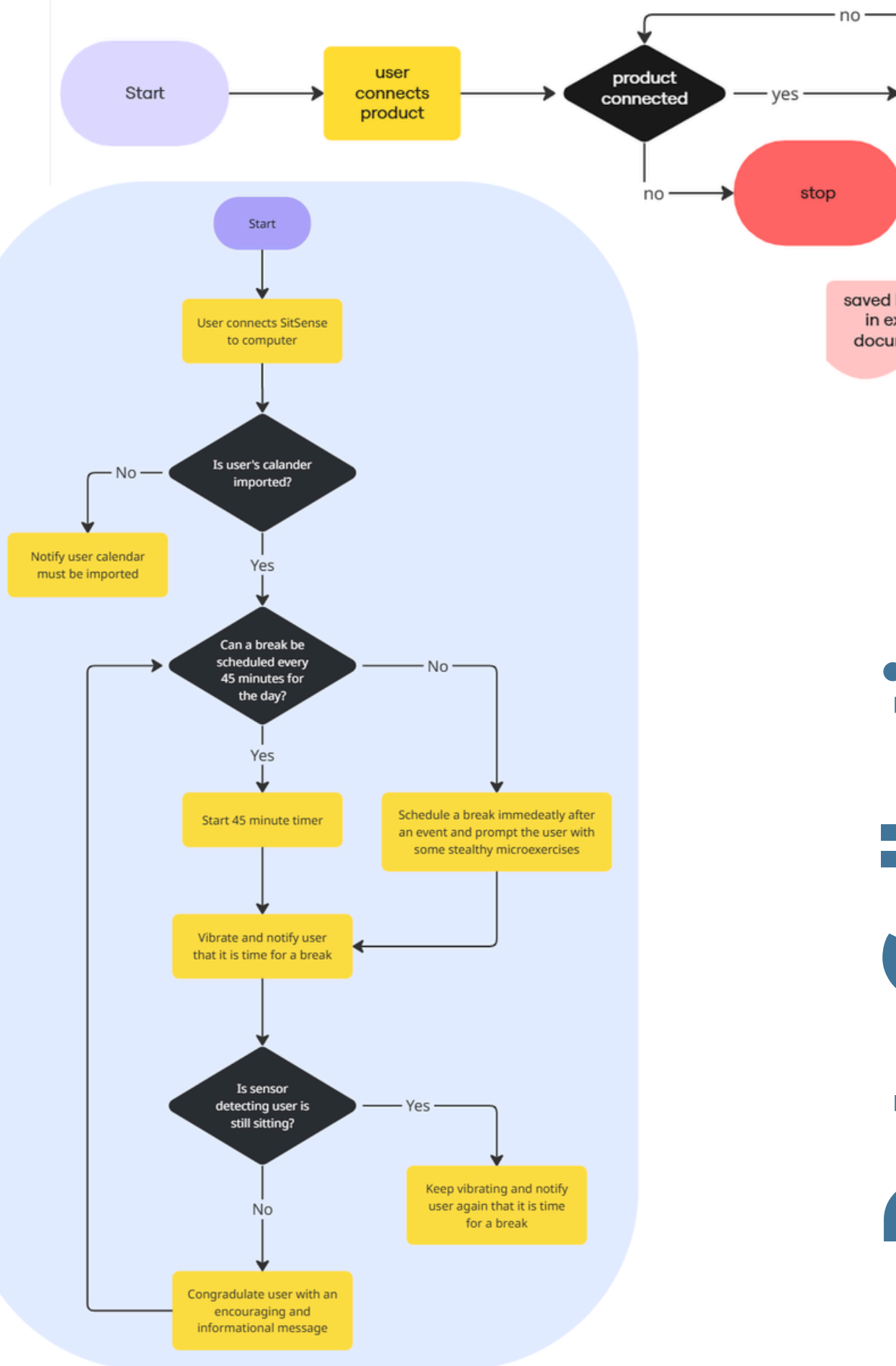
Sitsense is an ergonomic pillow used to help users increase their motivation through user experiences. To make it able to be used for a long period of time, it is key to use good cushioning and ergonomic shaping. Looking at the market, there is already a good amount of ergonomic pillows that help users have a good seating position during work. Our design goes a step further, including smart elements, such as limit switches inside the pillow that detect the use of the pillow, vibration motors in the pillow to nudge the user to stand up and take breaks in between work and lastly an app for computers, to set timers, see progress and get feedback on their use with the product.

The seat has a layer of soft cushioning, giving enough comfort while using, but also gives an opportunity for the product to detect the weight and use of the product through the limit switch underneath. To increase the user's interaction and experiences with the product, we made it so that the product only works when a user sits on it, making them connect to their own product with intent.

The buzzers are used to inform users to take active breaks, by nudging them in an uncomfortable way. This is not optimal, as it might influence the experience of the user in a bad way, but we have used multiple other techniques in the service to make up for this potential loss, to make the user feel more connected to the product. The use of variety in pillow colors, ability to plan working hours or breaks themselves, and the overall simplicity of the product, so that it is integrated into their surroundings, boosting the effectiveness of the product experiences.



Data Collection



Sitsenses' smart thinking process, differentiates it from the already existing products regarding micro-exercise breaks. To collect data, new methods are implemented, that will improve the connection and user experience throughout the use of the Sittsense.

Sitsense gives the opportunity for the user to upload their personal selected Google Calendar. This provides satisfactory benefits for the user, as they feel more integrated into the system of the product by having their own input, even though the algorithm of the app will decide when breaks should happen, according the 45 minute timer. Next to it is the limit switch inside the ergonomic pillow, which is used to measure the actual interaction of the user taking part of the user experience, sitting down when the timer is active and standing up when the timer runs out. When the Sitsense detects the user standing up to take a break, another timer will run, this time to collect the time the product is not in use. When the user comes back from their break, this time is stored in an Excel document, stored locally on the device, which is used by the interface to collect and give feedback upon.

INTERACTION CONCEPT & DATA COMMUNICATION

Introduction / Interaction Concept

SitSense connects a seat-mounted clicker sensor with a desktop interface that encourages short micro-exercises and stress-relief breaks.

The interaction concept is reactive: it responds in real time to idleness data and the user's digital calendar. When prolonged sitting is detected and no meeting is scheduled, a gentle vibration and short on-screen message invite the user to move. The design follows the principles of simplicity, transparency, and responsiveness. Every signal has a clear cause, and users can control when nudges occur through calendar settings. By keeping feedback local and tangible, SitSense supports user autonomy and competence, users feel guided, not monitored, in forming healthier habits.

How the Interface Works

After installation, users pair the cushion sensor and sync their calendar. The app runs quietly in the background, tracking sitting time and skipping nudges during meetings. When the 45-minute threshold is reached, the seat vibrates and a friendly message appears such as:

"Did you know short movement breaks boost focus and reduce stress?"

When the user stands, the break is logged and the dashboard offers brief positive reinforcement.

The interface relies on clear visual hierarchy, simple icons, minimal text, and timeline of user's micro-breaks, keeping attention on feedback rather than settings. This approach uses positive reinforcement, following the idea that behaviour change should be encouraging and unobtrusive.

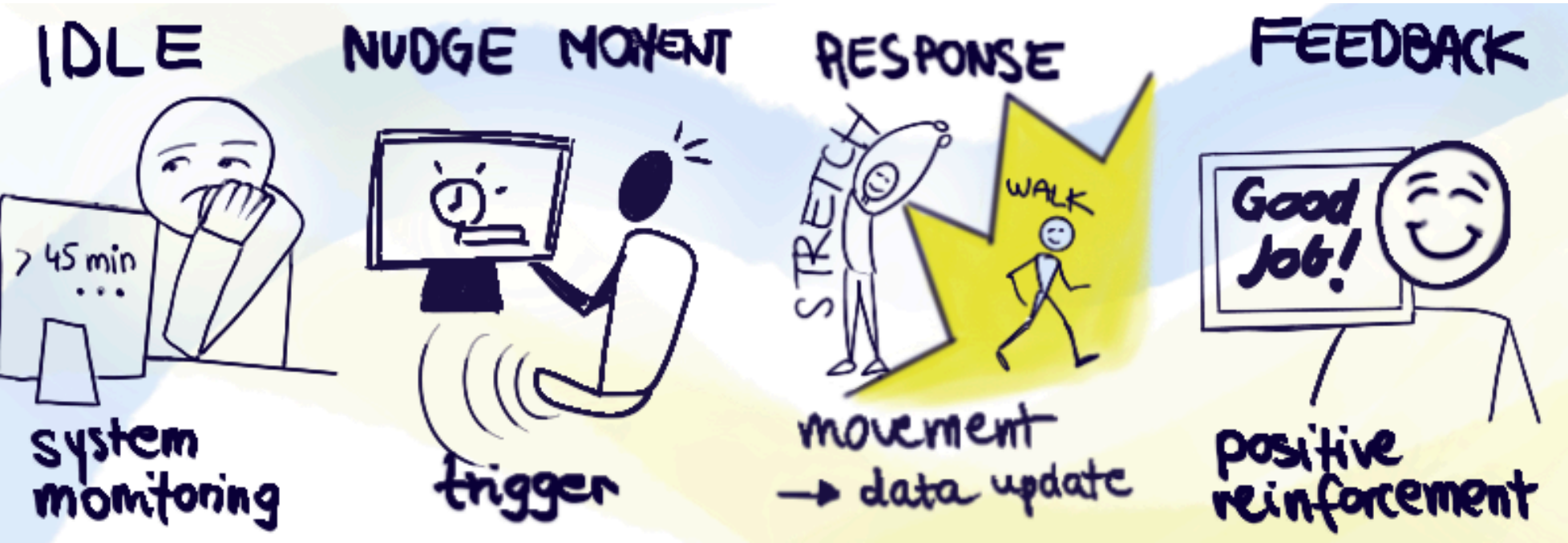
Data Communication

SitSense shows only what matters: time spent sitting and breaks taken. Progress appears in compact bar or ring charts and short encouraging messages.

Data updates automatically and is stored locally to maintain privacy and trust. The system is data-informed: it uses information to shape meaningful feedback rather than dictate user behaviour.

Data Enables Action

SitSense translates raw data into gentle, actionable cues. When idleness is detected, vibration and on-screen feedback prompt movement; once movement is sensed, the action is confirmed visually and logged. This closes the loop:
data → nudge → action → confirmation → record,
creating an easy rhythm of awareness and reward. Local storage reinforces ethical, transparent data use.



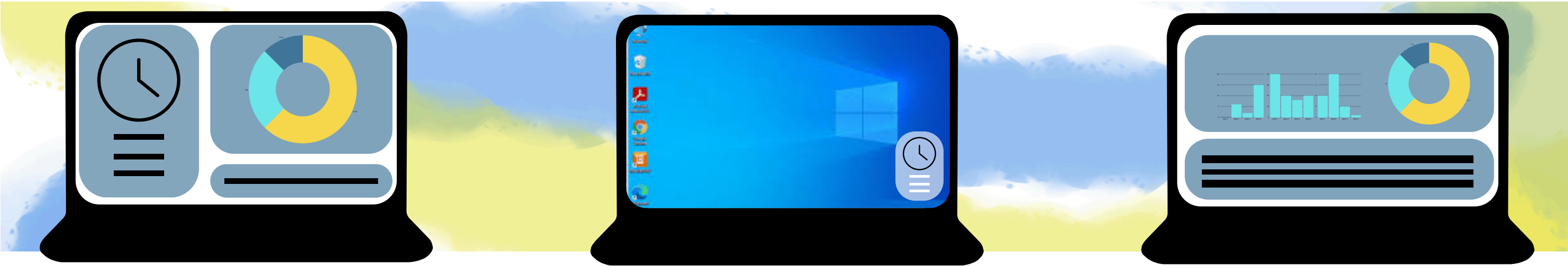
Feedback, States & Uncertainty

Three simple states structure feedback: Active, Idle, Break taken. The system reacts instantly to sensor input, confirming that each action has been registered. With no loading or error screens, interaction feels continuous and calm. The design is intentionally unobtrusive and controllable, avoiding pressure or disruption while still keeping users aware of their activity.

Reflection / Insight

Clear, immediate feedback proved more effective than persuasive tactics. By making system states visible and supportive, SitSense encourages self-reflection and intrinsic motivation - users move because they choose to, not because they must.

Through its modest cues and honest data communication, SitSense turns passive information into a quiet partner for wellbeing, showing how small, transparent interactions can foster positive behavioural change.



PRIVACY - INTENDED APPROACH

For insuring privacy of the users while collecting data, an info brochure that informs the user about the consequences of the brochure that are taken, data types that were collected, why is is collected and how is it going to be processed. These steps all took place before testing the users.This brochure contained in-depth information about why is that study being excuted and what, how and why data are being collected. Participants could ask questions and withdraw freely, ensuring autonomy. Diverse age groups were included for more representative results. Demographic data (age, gender, occupation) were collected maily for statistical purposes and scheduled for deletion after the module to protect privacy.

Unintended consequences

However, our considerations were not flawless, as several aspects came up. Due to maintaining the privacy, the videos taken were kept anonymous without showing the face expressions of the users, which would have been useful in the process of collecting data. Secondly, the data collected were prone to be biased for awareness, since the users knew that they were being recorded, which unintentionally made their body aware of the fact that they are being tested so the results might have been less genuine

Context

During generating the microsoft forms, it was indicated that different age ranges would be included to make sure that the availability of trying out as many users as needed data are being collected. Therefore, it was also that the data with the most vulnerability (demographics) are collected for statistical reasons and would be deleted as soon as this module is over. In general, it was challenging to balance between the data richness, quality and ethical responsibility. This explains the hassle of choosing a wide range of age in the the form to be able to get alot of data but also this increases the responsibility of the privacy concerns.

H1G2

Ergomotion Chair - Enhancing micro-exercise through Nudges

About this project:

The Ergomotion Chair is an innovative seating solution designed to encourage micro-exercising

After 45 minutes of continuous sitting, the chair delivers a gentle, automated nudge that simulates micro-exercises movements that promote better circulations.

Special data:

We do not intentionally collect any of the following sensitive categories of personal data:

Only health data is partially relevant as it gives an indirect indicators such as posture comfort or movement response may be noted, but no medical information is collected.



Sub-Processing Under How We Handle Your Data

Your personal data will be handled according to the General Data Protection

Regulation (GDPR) principles. We collect and process only the data necessary for research purposes, as outlined:

Data type	purpose of processing	how it is processed
Basic demographic data (age, sitting frequency)	To understand user diversity and ergonomic suitability	Used only for statistical analysis; anonymized in reports.
Feedback data (survey responses, verbal feedback)	To assess usability and comfort experience	anonymized for analysis.
Interaction data (sitting duration, nudge response time)	To evaluate physical response and usability patterns	Data anonymously stored before analysis through microsoft forms

Data retention and security

- All personal and feedback data will be stored securely for a limited time (until the end of this module).
- Only authorized examiners and module coordinators will have access.
- Data will be deleted after the research concludes.

Your rights as a participant:

- Access, correct, or delete their data.
- Withdraw consent at any time.
- Request information on how their data is being used.

Testing

A microsoft forms was made of quantitive questions so acceptance test of the prototype can be made according to the users preferences and answers.

- showing visuals of graphs and results (when some results are generated)

journey map | Sit sense



office worker/
student

important characteristics

commutes by car/public transport
sits most of the day

most important goals of the profile (in this scenario)

1. increasing the amount of physical activity
2. raising awareness on the benefits of such
3. implementing the product in daily (work)

key insights

Most important findings in this journey map

1. Immediate, transparent feedback is more effective than persuasive tactics.
2. Local, minimal data builds trust and autonomy.
3. Small, unobtrusive cues drive consistent behaviour change better than heavy gamification.

system actions

awareness

user learns about the app as a solution to their sedentary lifestyle

App promoted as a solution to sedentary work habits; simple demo shows vibration cue and gentle notifications.

onboarding

user downloads app, connects the cushion to their chair, imports calendar

System checks if calendar is imported and confirms successful pairing.

learning

user snoozes/ accepts nudges

App quietly monitors sitting time; after 45 minutes, cushion vibrates and screen displays a short nudge ("Time for a quick

practice

user regularly responds to reminders, participates in active micro-breaks

System detects standing through sensor and congratulates user with positive message.

mastering

user forms a habit of breaks

Daily dashboard shows sitting vs active time; system adapts timing based on schedule.

experience



intrigued

curious, hopeful, but
skeptical

learning, adjusting

proud, productive

empowered, satisfied, can be
bored at times

needs

- understand why micro-breaks matter
- see credibility and simplicity of use

- hassle-free installation
- clarity on what data is used
- immediate feedback that the system works

- understand timing logic and control notifications
- positive reinforcement for first interactions

- consistent, non-disruptive feedback
- confirmation that breaks are tracked correctly

- long-term motivation without pressure
- clear record of improvement

opportunities

- encourage to get started, raise awareness

- streamline first setup (automatic detection, guided steps)
- highlight privacy and transparency to build trust

- offer light customization (snooze, delay)
- provide short explanations of benefits after breaks

- reinforce good habits through short stats or visuals
- keep tone friendly, never pushy

- celebrate milestones visually
- suggest gentle long-term goals (e.g., consistent weekly activity balance)

TECHNOLOGY MODEL AND USER ACCEPTANCE TEST

The technology acceptance model was crucial in our prototype to assess user acceptance based on perceived usefulness and percieved ease of use.

Perceived usefulness was the degree to which our users believed the system of the padding and the user interface in the chair enhanced their job performance through activating the micro-exercise.

Perceived Ease of Use is the degree to which the users believe that the system of control is easy to understand and operate.

Attitude Toward Use (ATU) is the behaviour of the testing user against the use of the product.

Findings and interperitition:

Users showed divided opinions throughout the questions in the survey that covered ease of use, attitude towards use and the percieved usefulness.

PU is moderate, suggesting users recognize the technology's potential but do not yet see strong, general productivity benefits. In the context of the behavioural intention (BI), Data interperitition

- 11 users will use ergomotion if available.
- 3 intend to use it actively.
- 1 is not planning to use it.
- 0 would not recommend it.

Behavioural intention is high, with approximately 82% showing willingness to use the product. This indicates strong adoption potential once usefulness and usability are optimized.

Perceived Ease of Use (PEOU)

- Average Rating: 3.93 / 5

Distribution: Most ratings between 3 and 5.

Interpretation:

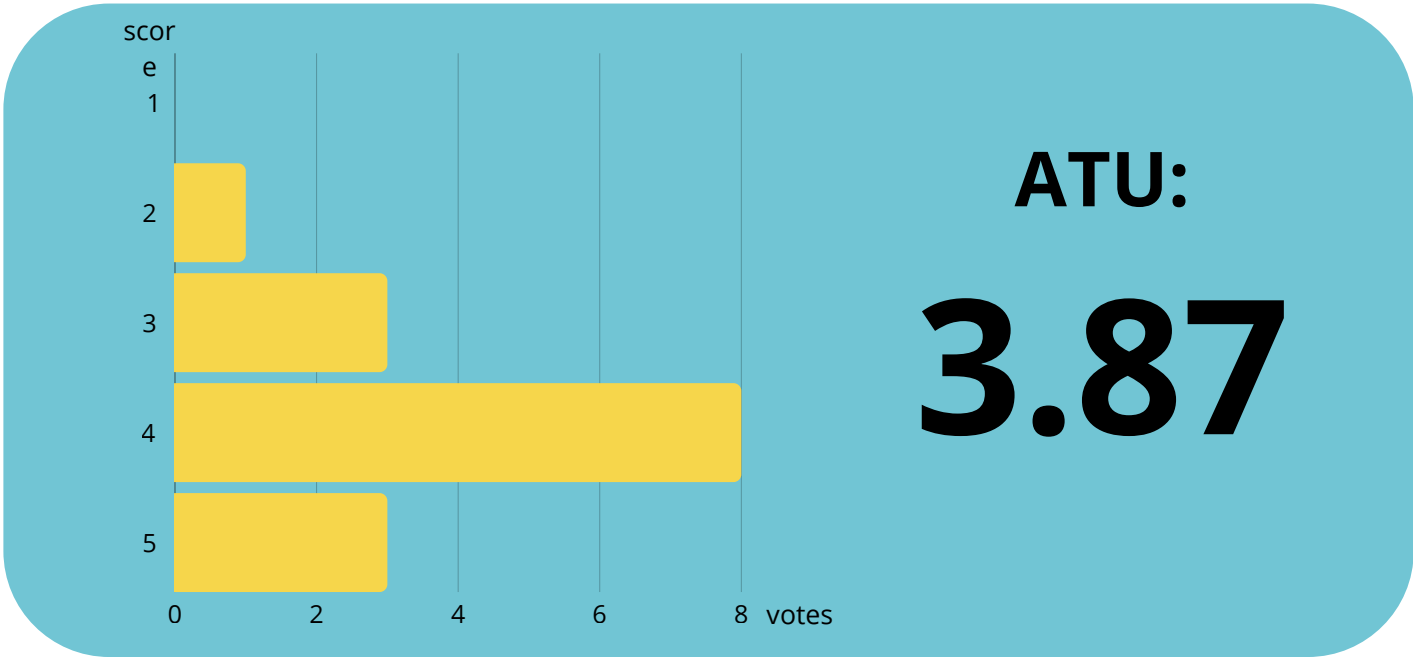
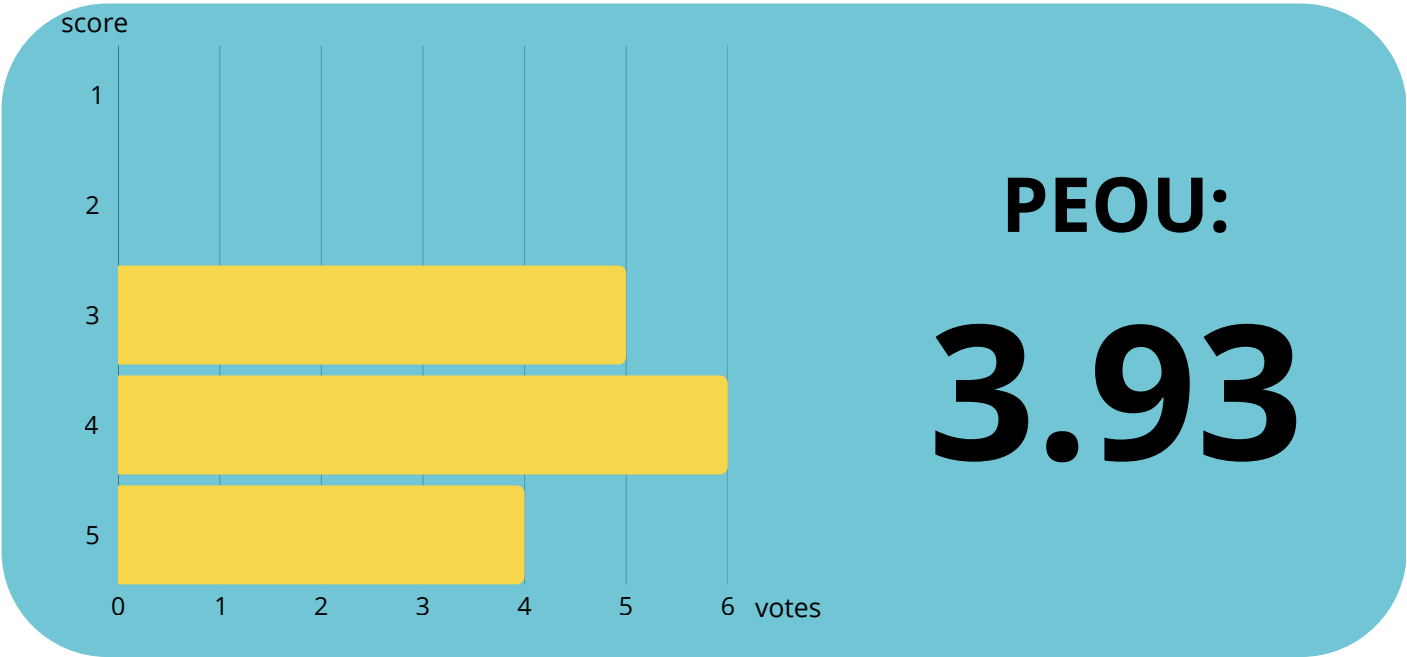
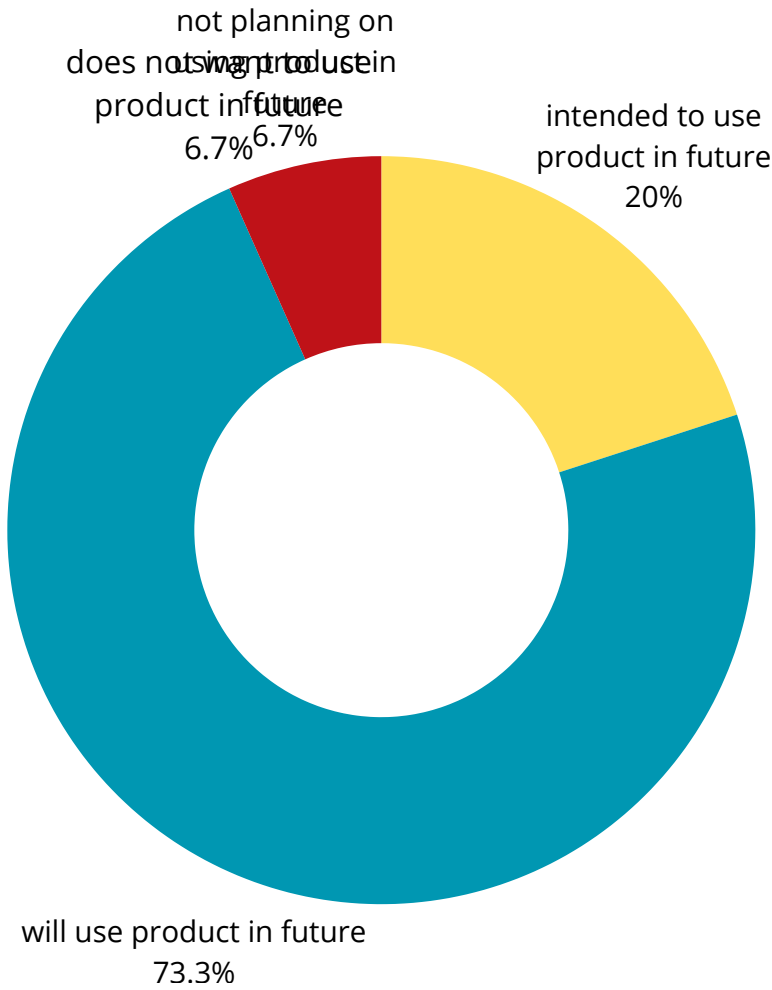
Precieved usefulness is above average, suggesting that users generally find the chair and interface easy to use. However, minor usability improvements (interface clarity, feedback cues, or onboarding) could raise satisfaction further. Attitude Toward Use (ATU)

- Average Rating: 3.87 / 5

Distribution: Majority rated 4 or 5, indicating positive emotional response.

Conclusion:

The results show that most users have a positive attitude towards the product and system with a willingness to use it in the future. Although the perceived usefulness and ease of use are above average, improving the interface and guidance could increase the overall user satisfaction.



FUTURE IMPROVEMENTS:

The user feedback provided some valuable insight into some potential changes of the prototype. While the initial design successfully demonstrated the feasibility of combining ergonomic support with interactive technology, some areas for refinement were identified to enhance usability, personalization, and user experience.

Participants suggested simplifying user control. Ideas such as a mobile app interface, as opposed to one on a computer would make it more convenient and accessible. Additionally, some suggested linking the user's schedule by using a sync feature or by connecting their account through an app setting. Though this would imply more privacy and data handling concerns, it would increase usability of the product as it would increase the efficiency of connecting the user's calendars, this would only need to be done once and it would update automatically if any changes are to be made, without the user needing to download and upload a new .ics file of their calendar every time a change is made.

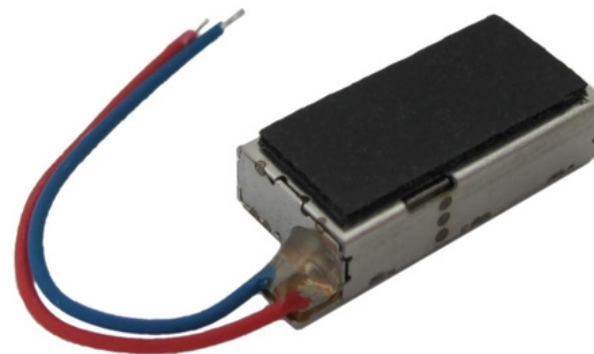
From a usability standpoint, the Technology Acceptance Model (TAM) results already indicated good perceived ease of use, but future iterations could benefit from clearer visual cues in the interface and a smoother onboarding experience for first-time users.

Due to the time and budget constraints while working on the prototype, a buzzer was used to notify the user when it is time to take a break. The final product would use multiple vibration motors, such as ERM coin motors or LRA motors. With regards to wiring, they would be connected to separate GPIO pins to allow for independent control so that different vibration patterns can be created. Additionally, microcontrollers (such as Raspberry Pis or Arduinos), can only supply a small amount of current from their output pins, typically 20-40 mA per pin, and often at 3.3 V or 5 V, however even tiny vibration motors draw 70-150 mA each. Trying to connect the motors directly to the GPIO pins could cause damage to the pins or the whole microcontroller itself. Using a MOSFET as an electronic switch would allow for safe control of higher-current devices like these motors.

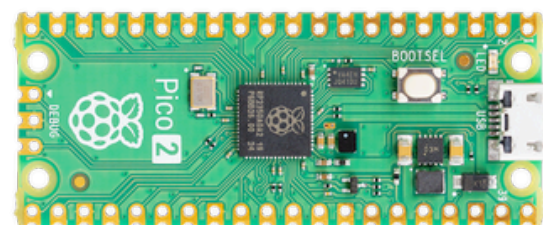
By incorporating these user-driven insights, future developments of the system could create a more engaging, adaptable, and user-centered ergonomic experience, ultimately increasing both acceptance and long-term adherence.



200hz LRA Vibration Motor



200hz LRA Vibration Motor



Raspberry Pi Pico W2

DISCUSSION AND CONCLUSION:

The research discussed the disadvantage of the lack of the micro-exercises as it can lead to cardio-vascular health risks. Behavioural science suggested that small exercises would decrease the risk of the consequent heart diseases they get.

The step of aligning a product that works as a solution for the problem was done through the product architecture that showed main features like sensor detection, in which detects the standing and the sitting of the user. It also showed the timer system which described the 45 minutes interval. Other features in the product architecture helped to understand what features should be included in the product like feedback loop.

This matches with the research by addressing the sedentary movements, encourages behavioural change through nudging, as well as supports autonomy and intrinsic motivation.

The real-time response to idleness and motion through vibration and on-screen prompts was the main part found with the data communication. Through simple icons, and mainly communicating the main data and storing it privately.

Taking the research done as an initial step in consideration, we carefully thought that the way to communicate our data worked as a gentle cue and a simple feedback to lower your stress instead of having additional digital fatigue. Moreover, it emphasized gaining user trust and using autonomy.

The prototype significantly showed the potential of integrating ergonomics into interactive technology, to promote micro-exercise.

Overall the (TAM) indicated that the users generally find it easy to use and moderately useful.

References

- Johnston, P. (2018). Demystifying Microcontroller GPIO Settings. [online] Embedded Artistry. Available at: <https://embeddedartistry.com/blog/2018/06/04/demystifying-microcontroller-gpio-settings/>. <https://www.vybrionics.com/coin-vibration-motors/with-brushes/v-c1030b028f>
- ProductInfo. (2020). Guide to MOSFETs. [online] Available at: <https://spencekonde.github.io/ProductInfo/MOSFETs/Guide.html>
- MIKROE. (2025). Vibration ERM Motor 9K RPM. [online] Available at: https://www.mikroe.com/vibration-erm-motor-9k-rpm-3v?srsId=AfmBOoqRUFL0-vtOlirS-lzVyw8uAVhAUYWkTPiufxLxjwrgf_2Zesal [Accessed 7 Nov. 2025]. <https://www.vybrionics.com/linear-lra-vibration-motors/v-l120628h>
- Andersen, L. L., Skovlund, S. V., Vinstrup, J., Geisler, N., Sørensen, S. I., Thorsen, S. V., & Sundstrup, E. (2022). Potential of micro-exercise to prevent long-term sickness absence in the general working population: prospective cohort study with register follow-up. *Scientific Reports*, 12(1), 2280. <https://doi.org/10.1038/s41598-022-06283-8>
- Balci, R., & Aghazadeh, F. (2004). Effects of exercise breaks on performance, muscular load, and perceived discomfort in data entry and cognitive tasks. *Computers & Industrial Engineering*, 46(3), 399–411. <https://doi.org/10.1016/j.cie.2004.01.003>

- Silva, A. P., Prado, S. O. S., Scardovelli, T. A., Boschi, S. R. M. S., Campos, L. C., & Frère, A. F. (2015). Measurement of the Effect of Physical Exercise on the Concentration of Individuals with ADHD. PLOS ONE, 10(3), e0122119.
<https://doi.org/10.1371/journal.pone.0122119> .
- Utwente.nl. (2025). Redirecting. [online] Available at:
<https://canvas.utwente.nl/courses/17281/pages/activity-overview-data-driven-product-development-and-interaction-including-lecture-slides>.
- Batista-Ferreira, L., Sandy, D. D., Silva, P. C. M. C., Medeiros-Lima, D. J. M., & Rodrigues, B. M. (2024). Impact of active breaks on sedentary behavior and perception of productivity in office workers. Revista Brasileira de Medicina Do Trabalho, 22(2), 1–9.
<https://doi.org/10.47626/1679-4435-2023-1213>
- Hossam El-Zalabany. (2024). Mobile notifications as an information medium: an ethnographic study of mobile human-information interaction attitudes of Egyptian and German higher education students. Cybrarians Journal, 71, 1–33.
<https://doi.org/10.70000/cj.2024.71.585>